Collaborative Response of Data Center Coupled With Hydrogen Storage System for Renewable Energy Absorption

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***Abstract*—The demand response of data center is considered as an effective flexible method to absorb excessive renewable energy (RE) in the power system, especially for the data center with high-density hydrogen storages. Therefore, this article proposes a collaborative response framework considering regulations of data workload and hydrogen storage. It is expected to further improve the data center flexibility. However, this collaborative response is a high-order and non-convex optimization problem involving the multi-period and complex model. Considering the condition of uncertain RE supply, it is difficult for power optimization dispatch to realize the precise decision in this NP-hard problem. To address this issue, a global interval optimization (GIO) approach is developed. It is to obtain efficient and robust decisions in the collaborative response. This approach deduces an interval optimization model to achieve the global optimality of convergent solution. Furthermore, this article introduces a probability correction into the GIO approach. It is used for improving the decision accuracy in the actual renewable power scenarios. A case study based on the IEEE 24-bus system verifies superiority of the GIO approach. It further validates effectiveness of the collaborative response framework for RE utilization.**

***Index Terms*—Collaborative response, data center, global interval optimization, hydrogen storage system.**

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I. Introduction

A. Backgrounds

TO REPLACE the conventional thermal power on the supply side in power systems, the integration of high-capacity renewable energy, such as wind power (WP), is considered to be one of the effective methods. This transformation not only addresses the unsustainable problem of energy supply, but also reduces carbon emissions caused by the burning of fossil energy [1], [2]. However, there is a large WP available supply during the valley period of power load commonly, which is called the WP anti-peak characteristic (WPAPC). With the increase of grid-connected WP, WPAPC will be more notable, which further aggravates the difficulty of WP utilization. Therefore, it is necessary to use corresponding feasible methods to absorb the excess WP during the valley period [3], [4].

An effective idea to adjust the demand of large power customers to promote WP absorption and alleviate the WPAPC [5]. Especially for the data center, a large-scale electricity infrastructure, the power demand adjustment is highly centralized and automated [6]. It mainly involves: (*i*) the management of power load; (*ii*) the management configured energy system, such as diesel generator and battery storage system.

There are many related works that concentrate on load management and renewable energy absorption. For instance, Ref. [7] studies the load migration of data centers from PJM to CAISO, two independent regional operators of the US power system. It leads to the effective result for promoting the utilization of renewable energy and reducing carbon emission. Among multiple data centers in region, Ref. [8] adopts a randomized strategy to schedule the migration of data workload, in order to achieve the increase of renewable energy utilization in the grid. For studying the management of internal consumption, Ref. [9] formulates a power model of data center to quantify load variation. This model involves the adjustment of the working frequency in the information technology (IT) servers and the geographic migration of the data workload. The above works mainly focus on achieving power demand shifting based on the centralized scheduling of IT servers and data workload.

On the other hand, the original purpose of energy system management is to stably supply power to data centers in emergencies.

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Nevertheless, the diesel combustion will generate environmental contaminants, as well as cause the increase in carbon emission. For conventional battery storage, its energy capacity is limited by the position of the over-size device [10]. Moreover, the frequent operation may lead to rapid attenuation of battery capacity [11]. Therefore, existing energy management methods are not effective for the clean and steady power supply to data centers.

The development of hydrogen storage systems (HSS) may address the above dilemmas [12]. The clean reaction product, water, will not cause pollution [13]. High-density hydrogen energy occupies less area, which guarantees equipment configuration of large-scale capacity [14]. Furthermore, it is worth noting that extended storage capacity provides the potential to store excess WP. Specifically, Ref. [15] studies the bidirectional conversion, power-to-hydrogen (P2H) in the electric tank and hydrogen-to-power (H2P) in the fuel cell, to store intermittent renewable energy and address unpredictable load fluctuation. In Ref. [16], a rule-based microgrid operating strategy is proposed to analyze the HSS's positive role in stabilizing microgrid operations while using high-proportion renewable energy. In recent years, more data center operators deploy large-scale HSSs to ensure power supply reliability [17]. For instance, Microsoft data center proposes the green and sustainable project: alternation of the existing diesel generators by HSSs and the objective of negative carbon emission [18]. Based on the expectable advancement, it is a valuable idea in data center operating to comprehensively apply power load management and existing HSS systems to utilize WP and alleviate the WPAPC, which is also one of the concerned research contents in this article.

Apart from WPAPC, the uncertainty of WP supply is another dilemma for its effective utilization [19]. This is reflected in the adverse impact of fluctuating WP on the power supply-demand balance and transmission branch. These characteristics will degrade the stability and security of the power system. Various research works concentrate on achieving power dispatch in the situation of WP uncertainty. The main methods can be divided into probabilistic distribution-based stochastic optimization and robust optimization. Nevertheless, in the specific scenario, stochastic optimization depends on a large number of scenario data to model reliable distribution. With the increase of widely distributed WP scenarios, data sampling and distribution modeling will be difficult to achieve. In addition, the given distribution model is insufficient for considering extreme conditions in uncertain optimization [20]. On the other hand, robust optimization searches the worst-case conditions, i.e., pessimistic scenarios, in the uncertain set to determine the corresponding decision [21]. It causes the robust solution which usually tends to be conservative.

B. Related Works

A strategy called interval optimization (IO) is proposed to address the uncertain optimization problem [22]. In decision-making, this strategy considers not only the robustness involving the extremely pessimistic scenario but also the efficiency involving the optimistic scenarios, i.e., the best-case conditions. Contrast to the above mentioned robust solution, the IO solution set is capacity to avoid overly conservative results. Furthermore, IO only depends on the fluctuating boundary of uncertain variables. This available boundary is easier to obtain compared with a large number of accurate probability distributions [23].

Some works apply the IO strategy in different power-dispatching situations to achieve decision-making within uncertainties. For instance, Ref. [24] formulates a linear interval model to improve the coordination of uncertain WP and hydropower. This linear IO model achieves the weighted bi-objective optimization process, based on the arithmetic rules of linear interval functions. On this foundation, Ref. [25] establishes a dual interval linear optimization model in the day-ahead contract electricity market to optimize transactions involving uncertain factors. This dual IO is decomposed into several simple sub-models in the solving procedure. In fact, for linear optimization, optimistic and pessimistic scenarios are proved to exist at the interval boundary. This nature is the premise of relaxation and realization of linear IO. However, for high-order and non-convex problems, this simple explicit property is not tenable. Therefore, in order to realize high-order and non-convex IO, it is necessary to study how to obtain corresponding pessimistic and optimistic scenarios. The purpose is to evaluate the effectiveness and robustness of decisions, based on the pessimistic and optimistic value.

In this context, for the optimization model containing 0–1 variables, Ref. [26] decomposes the IO model of the thermal unit commitment into main optimization and the sub-problem of the interval model through Bender decomposition. This sub-problem is only used to select interval feasible solutions in the iteration. The method of decomposition and iteration essentially requires an exhaustive enumeration of all scenarios in the interval to evaluate the decision. To reduce the computational load, a scenario reduction strategy is applied according to the specific constraint properties in this problem. However, this idea is subjective and difficult in wide application. Ref. [27] formulates the process of generating pessimistic and optimistic scenarios and corresponding decision-making as a bi-level optimization. On this basis, this work uses an evolutionary algorithm to achieve this bi-level relationship and optimal power flow. However, this algorithm also causes a heavy computational load, and cannot guarantee the convergence and optimality of its solution set. Ref. [28] introduces robust optimization as a sub-problem to characterize pessimistic scenarios, and develops a decomposition and alternative algorithm based on fixed integer. The purpose is to obtain optimistic and pessimistic scenarios. This algorithm only aims to obtain the two scenarios rather than specific decisions. Although the convergence of scenarios is proved based on the gradient trend of the *Lagrange* function in iteration, the optimality of the decisions corresponding to these scenarios cannot be guaranteed.

Based on the above literature review, aiming at the flexible scheme of power dispatch and the uncertain IO approach, we summarize the following research gaps.

1) The collaborative response of the data center coupled with hydrogen storage system is a novel and feasible research topic, which is able to not only reduce the external power dependence for data centers, but also boost the absorption

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of excess WP. It is a demand response scheme worthy of research and application, especially in power system dispatch considering WP uncertainty.

2) An interval optimization approach suitable for the high-order non-convex problem of the above collaborative response is lacking. The existing methods are only applicable to specific models, and cannot achieve the optimality of the convergent solution. Some attempts involving evolutionary algorithms are challenging to address the problems of convergence and optimality, leading to excessive computing demand.
3) The traditional interval optimization model ignores the differentiated distribution of scenario probability. Uncertain factors only follow boundary constraints. This means that the scenario distribution is defaulted as uniform. If the interval decision is optimized without probability correction, the actual effectiveness and robustness of the optimal decision would be weakened.

C. Contributions

In our work, the load management of data centers and the application of affiliated hydrogen storage systems are considered as the extension of the power dispatching scheme. To address the fluctuation of WP supply, the high-order and non-convex IO model is established to characterize the power dispatch in uncertainty, and an applicable IO approach is proposed to achieve the globally optimal decision. Furthermore, a correction ideal of extreme scenarios is introduced into the IO model. It increases the probability correction to pessimistic and optimistic scenarios, and makes the decision more accurate in the actual distribution.

1) The collaborative responding framework of the data center coupled with hydrogen storage system (DCCHSS) is proposed and formulated as the high-order and non-convex interval optimization (HNIO) model. Its flexibility is used to address the anti-peak characteristic and uncertain fluctuation of WP supply.
2) The global interval optimization (GIO) approach is developed to achieve the DCCHSS scheduling in uncertain conditions and obtain the efficient and robust decisions. Based on the corresponding sufficiency condition, this approach is able to address the above HNIO problem by the complete constrained optimization model to optimize the global optimal solution set.
3) The standard deviation degree as probability correction is introduced into GIO. It makes pessimistic and optimistic scenarios more high-probability and improves the accuracy of GIO decisions.

The rest of this article is organized as follows. Section II establishes the DCCHSS collaborative responding model in the regional power system structure. Section III introduces the interval optimization model considering probability correction. Section IV develops a global optimal approach to achieve the above optimization idea. Section V focuses on the case study and analyzes simulation results. Finally, conclusions are drawn in Section VI.

II. Multiple Data Center Collaborative Response Considering Hydrogen Storage System

A. Power Dispatching and Demand Response

The main limitations of WP absorption can be attributed to the following aspects. (*i*) The WP supply in peak-period sometime is incompatible with power demand. WPAPC causes oversupply and grid-connected WP has to be reduced. (*ii*) The power capacity of the transmission branches limits the large-scale and centralized WP connection. Especially when the power demand is increased to utilize WP, the branches in some areas may be overloaded.

In this article, a collaborative response of the DCCHSS operation management is proposed by taking advantage of a flexible dispatch method to adjust the power demand in datacenter-connected buses during each period. In addition to emergency power supply, HSS is able to be applied to store excess WP in the WP peak periods or reduce the dependence of data centers on external power in the WP valley periods. Considering the coupling relationship between load and HSS, this response is a novel flexible method for power dispatch. Specifically, as shown in Fig. 1, geographically dispersed data centers are the flexible power load. In the WP peak periods, the IT operating load and hydrogen storage load are increased to absorb excess WP. In the WP valley periods, the IT operating load is decreased, and hydrogen fuel cell supply electricity to reduce the dependence from data center on the external grid. In addition, the migration of data workload among data centers optical fiber is able to balance the multi-regional load and avoid branch overload caused by excessive load [29]. It is also conducive to WP utilization.

In addition, the corresponding thermal power generation and grid operation are also considered in power dispatch. In Fig. 1, the regional power system includes the transmission network, thermal plants, wind farm, traditional load, data center load and hydrogen storage system. For the uncertainty of WP supply, this article proposes a global interval optimization approach in the power dispatching field to take advantage of demand-side flexible resources to achieve an efficient and a robust scheduling.

B. Modeling of Data Center Load and Hydrogen Storage

In order to achieve the effective cooperative response, operators need to organize the IT server operation status and appropriate data workload distribution. Besides, the limits of grid operation are necessary to be included in the scheduling. These complex coupling relationships are fully considered in this collaborative load model, which is also the basis for implementing the novel flexible dispatch method.

(*i*) Model of IT server load

The power load generated by IT servers is divided into two parts. One is the static power P_{st}^{sv} in operation. The other is the dynamic power caused by numerical operation in the central processing unit (CPU). The dynamic power is related to the amount of workload and the working frequency in IT servers. In this context, the total power $P_{i,t}^{sv}$ of the IT server is formulated as (1) [30].

$$P_{i,t}^{sv} = X_{i,t}^{sv} \cdot P_{\mathrm{st}}^{sv} + k \cdot f_{i,t}^{3} \cdot \lambda_{i,t} \cdot (\mu_{i,t}^{sv})^{-1}, \tag{1}$$

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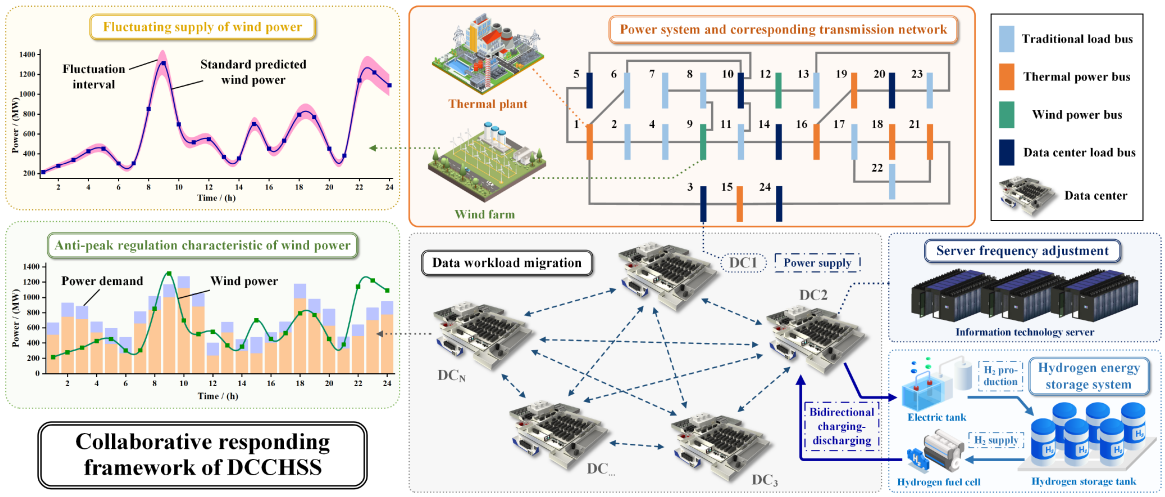


Fig. 1. Collaborative responding framework of DCCHSS.

where subscript $[\cdot]_{i,t}$ denotes the ith data center and tth period; $X_{i,t}^{sv}$ denotes the number of servers (sv); k represents the factor of dynamic power; $f_{i,t}$ represents the overall working frequency in all the IT servers; $\lambda_{i,t}$ is the workload in the data center operation; $\mu_{i,t}^{sv}$ stands for the overall service rate in all the IT servers.

The working frequency in the single IT server is an adjustable discrete option [9]. However, as the average of the working frequency in all the IT servers, $f_{i,t}$ is able to be considered the adjustable continuous variables in the allowable boundary. For service rate, it characterizes the amount of data workload processed in unit time, and has a similar physical concept with working frequency. Thus, there should be a proportional relationship between the two. On this basis, corresponding boundary constraints and equality constraints are formulated as (2).

$$f_{i,t}^{\min} \le f_{i,t} \le f_{i,t}^{\max}, \quad \mu_{i,t}^{sv} = \kappa f_{i,t}\,, \tag{2}$$

where $f_{i,t}^{\min}$ and $f_{i,t}^{\max}$ represent the lower and upper bounds of working frequency, and κ is a scale factor. The responding time for processing data workload consists of waiting time $T_{i,t}^{\mathrm{Q}}$ and operating time $T_{i,t}^{\mathrm{H}}$. The waiting time is depending on the redundancy ($\mu_{i,t}^{sv}X_{i,t}^{sv} - \lambda_{i,t}$) of IT servers, and the operating time depends on the service rate. The corresponding mathematical relationship is shown in (3).

$$T_{i,t}^{\mathrm{Q}} = \frac{1}{\mu_{i,t}^{sv}X_{i,t}^{sv} - \lambda_{i,t}}, \quad T_{i,t}^{\mathrm{H}} = \frac{1}{\mu_{i,t}^{sv}}\,. \tag{3}$$

We consider maximum operating time $T_{i,t}^{\mathrm{H,max}}$, and set the upper bound constraint of responding time in (4).

$$T_{i,t}^{\mathrm{Q}} + T_{i,t}^{\mathrm{H,max}} \le \mathrm{D}\,, \tag{4}$$

where D represents the maximum responding time. Some data workloads can be migrated to another data center for calculation. In fact, the actual migration workload $\lambda_{i,t}^{\mathrm{MI}}$ should be less than the available migration workload $IT_{i,t}^{\mathrm{MT}}$ in the migrate-out process. $\lambda_{i,t}^{\mathrm{MI}}$ is a vector and negative value represents migrate-out. Furthermore, among multiple data centers, the number of inbound and outbound workloads should be balanced, i.e., the sum value is 0. Based on the above principles, the constraints of data workload can be formulated as (5). On the other hand, the total workloads in data centers consist of migration workload and existing workload, as presented in (6).

$$\lambda_{i,t}^{\mathrm{MI}} + IT_{i,t}^{\mathrm{MT}} \ge 0, \ \sum_{i=1}^{\mathrm{N}} \lambda_{i,t}^{\mathrm{MI}} = 0, \tag{5}$$

$$\lambda_{i,t} = \lambda_{i,t}^{\mathrm{MI}} + IT_{i,t}^{\mathrm{MT}} + IT_{i,t}^{\mathrm{RT}}\,, \tag{6}$$

where $IT_{i,t}^{\mathrm{MT}}$ and $IT_{i,t}^{\mathrm{RT}}$ represent migratable and real-time workload, and $\lambda_{i,t}^{\mathrm{MT}}$ denotes actual migration workload. N is the number of data center.

(*ii*) Model of traditional load in data center

The traditional load $P_{i,t}^{\mathrm{tr}}$ in date centers mainly includes common building loads such as cooling and lighting, which can be obtained by the established value of power utilization efficiency (PUE). Thus, the total load $P_{i,t}^{\mathrm{DC}}$ in data centers is formulated as (7).

$$P_{i,t}^{\mathrm{tr}} = (\mathrm{PUE} - 1) \cdot P_{i,t}^{sv}, \ P_{i,t}^{\mathrm{DC}} = \mathrm{PUE} \cdot P_{i,t}^{sv}. \tag{7}$$

(*iii*) Model of bidirectional hydrogen (H_2) storage

The HSS operation can be divided into charging and discharging processes, and the corresponding charging power $P_{i,t}^{\mathrm{Ch}}$ and discharging power $P_{i,t}^{\mathrm{Dis}}$ follow the boundary constraints as (8).

$$P_{i,t}^{\mathrm{Ch,min}} \le P_{i,t}^{\mathrm{Ch}} \le P_{i,t}^{\mathrm{Ch,max}}, P_{i,t}^{\mathrm{Dis,min}} \le P_{i,t}^{\mathrm{Dis}} \le P_{i,t}^{\mathrm{Dis,max}}. \tag{8}$$

The discharging power is not grid-connected, therefore the discharge power should be not higher than the power demand of the corresponding data center. Additionally, the charging and discharging processes cannot exist at the same time. These constraints are established as (9).

$$P_{i,t}^{\mathrm{Dis}} \le P_{i,t}^{\mathrm{DC}}, \quad P_{i,t}^{\mathrm{Ch}} \cdot P_{i,t}^{\mathrm{Dis}} = 0. \tag{9}$$

In the charging process, the HSS consumes electricity as a power load to produce storable H_2. In the discharging process, the hydrogen fuel cell consumes H_2 stored to supply electricity

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as a power source. These physical relationships can be formulated as (10).

$$Q_{i,t}^{\mathrm{H}_2(+)} = \eta^{\mathrm{el}} \cdot \rho \cdot P_{i,t}^{\mathrm{Ch}}, \quad Q_{i,t}^{\mathrm{H}_2(-)} = \mu \cdot P_{i,t}^{\mathrm{Dis}} / \eta^{\mathrm{f}}, \tag{10}$$

where $Q_{i,t}^{\mathrm{H}_2(+)}$ represents the capacity of H_2 produced by electrolysis; η^{f} is the electrolysis efficiency; ρ is the producting H_2 capacity of per unit power; $Q_{i,t}^{\mathrm{H}_2(-)}$ represents the capacity of H_2 used for power generation; and μ is the H_2 demand for generating per unit power. On this basis, the stored capacity $Q_{i,t}^{\mathrm{SH}_2}$ in the HSS follows constraint (11).

$$Q_{i,t}^{\mathrm{SH}_2} = Q_{i,t-1}^{\mathrm{SH}_2} + (\eta^{\mathrm{ch}} \cdot Q_{i,t}^{\mathrm{H}_2(+)} - Q_{i,t}^{\mathrm{H}_2(-)} / \eta^{\mathrm{dis}}) \cdot \Delta t, \tag{11}$$

where η^{dch} and η^{dis} are the stored efficiencies of H_2 in the charging and discharging process, and Δt is the time interval.

C. Modeling of Regional Power System

This article considers the power system model including grid branch transmission, grid-connected thermal plants and wind farms. Firstly, the power supply-demand balance is formulated as the constraints (12).

$$\sum_{i=1}^{\mathrm{N}} (P_{i,t}^{\mathrm{DC}} + P_{i,t}^{\mathrm{Ch}} - P_{i,t}^{\mathrm{Dis}}) + \sum_{j=1}^{\mathrm{M}} D_{j,t}^{\mathrm{Tr}} = \sum_{k=1}^{\mathrm{H}} P_{k,t}^{\mathrm{Th}} + \sum_{w=1}^{\mathrm{W}} P_{w,t}^{\mathrm{WP}}, \tag{12}$$

where subscript $[\cdot]_j$, $[\cdot]_k$ and $[\cdot]_w$ denote the jth grid bus, the kth thermal power unit and the w_{th} wind farms; M is the number of grid bus, H is the number of thermal units grid-connected; W is the number of wind farm grid-connected; $D_{j,t}^{\mathrm{Tr}}$ stands for the traditional load in buses, $P_{k,t}^{\mathrm{Th}}$ represents the power generated by thermal units; and $P_{w,t}^{\mathrm{WP}}$ is the connected WP. Secondly, the generation of thermal power units should follow the boundary and climbing constraints as (13).

$$P_{k,t}^{\mathrm{Th,min}} \le P_{k,t}^{\mathrm{Th}} \le P_{k,t}^{\mathrm{Th,max}}, r_k^{\mathrm{d}} \cdot \Delta t \le P_{k,t}^{\mathrm{Th}} - P_{k,t-1}^{\mathrm{Th}} \le r_k^{\mathrm{u}} \cdot \Delta t, \tag{13}$$

where $P_{k,t}^{\mathrm{Th,min}}$ and $P_{k,t}^{\mathrm{Th,max}}$ denote the lower and upper bound of thermal power, and r_k^{d} and r_k^{u} represent the lower and upper bound of the climbing rate. Furthermore, the grid-connected WP should also follow the boundary constraints. Herein, suppliable WP $P_{w,t}^{\mathrm{WSP}}$ is the upper bound for the grid-connected WP, and it is an uncertain variable. We assume that $P_{w,t}^{\mathrm{WSP}}$ follows an established interval, thus the above constraints can be formulated as (14).

$$P_{w,t}^{\mathrm{WP,min}} \le P_{w,t}^{\mathrm{WP}} \le P_{w,t}^{\mathrm{WSP}}, \ P_{w,t}^{\mathrm{WSP,min}} \le P_{w,t}^{\mathrm{WSP}} \le P_{w,t}^{\mathrm{WSP,max}}, \tag{14}$$

where $P_{w,t}^{\mathrm{WP,min}}$ is the lower bound of the grid-connected WP, while $P_{w,t}^{\mathrm{WSP,min}}$ and $P_{w,t}^{\mathrm{WSP,max}}$ represent the predictable lower and upper bound of the established interval of the suppliable WP. Equivalently, the actual WP connection follows an uncertain upper-bound IO constraint. In this article, a direct-current power flow model is established to consider transmission limits [31]. In this context, the inflow power $P_{m,t}^{\mathrm{G}}$ of bus m and transmission power $TP_{q,t}$ in branch q are represented as (15).

$$P_{m,t}^{\mathrm{G}} = D_{j,t}^{\mathrm{Tr}} + \sum_{i=1}^{\mathrm{N}} s_{i,m}^{\mathrm{DC}} (P_{i,t}^{\mathrm{DC}} + P_{i,t}^{\mathrm{Ch}} - P_{i,t}^{\mathrm{Dis}}) - \sum_{w=1}^{\mathrm{W}} s_{w,m}^{\mathrm{WP}} \cdot P_{w,t}^{\mathrm{WP}} + \sum_{n=1}^{\mathrm{M}} B_{mn}(\theta_m - \theta_n), \quad TP_{q,t} = P_{m,t}^{\mathrm{G}} - P_{n,t}^{\mathrm{G}}, \tag{15}$$

where B_{mn} is the susceptance between bus m and bus n, while θ_m and θ_n denote the voltage phase angle of bus m and bus n. When bus m connects the ith data center or wth wind farm, $s_{i,m}^{\mathrm{DC}} = 1$ or $s_{i,m}^{\mathrm{WP}} = 1$, otherwise $s_{i,m}^{\mathrm{DC}} = 0$ or $s_{i,m}^{\mathrm{WP}} = 0$, and these values represent connecting status of load or power source, rather than variables. A security coefficient C_{ASC} is defined to characterize the risk of power transmission. C_{ASC}, as presentd in (16), denotes the minimum capacity percentage of all the branches in all periods. Specifically, $TP_q^{\max} - |TP_{q,t}|$ represents the available margin, where power $TP_{q,t}$ is non-directed. When its minimum margin does not exceed the limit, the transmission system could be considered security.

$$C_{\mathrm{ASC}} = \min \left(\frac{TP_q^{\max} - |TP_{q,t}|}{TP_q^{\max}} \right) \times 100\%, \tag{16}$$

where $TP_q^{\max}$ is the maximum transmission power of branch q. Note, it is necessary to limit this minimum margin to ensure the transmission security, as shown in (17).

$$C_{\mathrm{ASC}}^{\min} \le C_{\mathrm{ASC}} \le C_{\mathrm{ASC}}^{\max}. \tag{17}$$

D. Interval Optimization Model of Collaborative Responding DCCHSS

Based on the proposed flexible load model, we further construct the interval optimization. This model takes advantage of the operating management in the multi-datacenter and affiliated HSSs to address the WPAPC and uncertainty. Specifically, we set the minimization of WP curtailment as the optimization objective, which is generated as the difference of the available and actual connecting WP, $P_{w,t}^{\mathrm{WSP}} - P_{w,t}^{\mathrm{WP}}$. The corresponding optimization model involves multifarious quadratic equality constraints and 0–1 variables, thus it is a higher-order and non-convex model. Employing uncertain interval variables, this interval model can be established as (18).

$$\begin{aligned} \min_x \quad & F(x,y) = \sum_{w=1}^{\mathrm{W}} (P_{w,t}^{\mathrm{WSP}} - P_{w,t}^{\mathrm{WP}}) \\ \text{s.t.} \quad & (1)\text{–}(17), \end{aligned} \tag{18}$$

where x represents the decision variable, $x = [f_{i,t}, \lambda_{i,t}^{\mathrm{MI}}, P_{i,t}^{\mathrm{Ch}}, P_{i,t}^{\mathrm{Dis}}, P_{k,t}^{\mathrm{Th}}, P_{w,t}^{\mathrm{WP}}]$, and y represents the interval variable, $y = [P_{w,t}^{\mathrm{WSP}}]$. The IO is purposed to obtain the accurate decision in uncertain conditions. A detailed IO model will be introduced in the next section.

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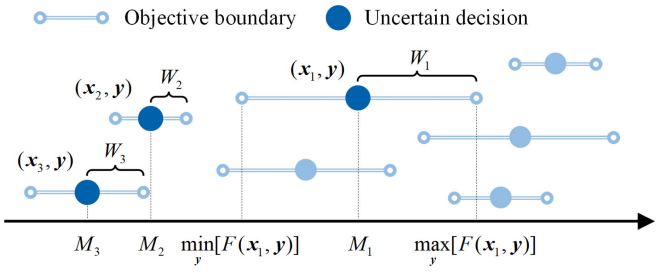


Fig. 2. Incomplete advantages of IO decision.

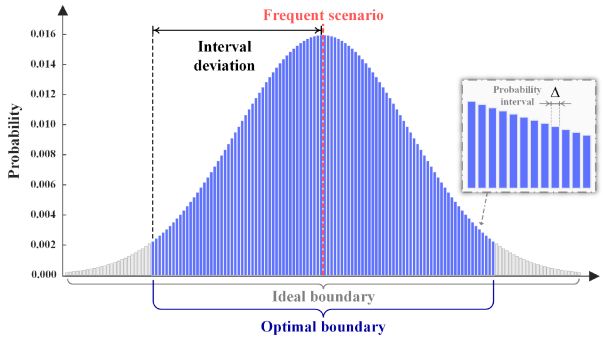


Fig. 3. Probability correction for pessimistic and optimistic scenarios.

III. Interval Optimization Considering Probability Correction of Pessimistic and Optimistic Scenario

From the perspective of decision-makers, they expect the optimal decision which can not only cause excellent results, but also have the stable effect on the changes in environmental conditions. Ref. [22] established two interval objectives to characterize the above optimization demand involving efficiency and robustness. Specifically, the median $M(x, y)$ and width $W(x, y)$ of the original objective $F(x, y)$ are shown as (19), (20).

$$M(x,y) = \left\{ \max_y[F(x,y)] + \min_y[F(x,y)] \right\}/2 \,, \tag{19}$$

$$W(x,y) = \left\{ \max_y[F(x,y)] - \min_y[F(x,y)] \right\}/2 \,, \tag{20}$$

where $\max_y[F(x, y)]$ denotes the objective value in the pessimistic scenarios, i.e., the worst result of the decision x. $\min_y[F(x, y)]$ denotes the objective value in the optimistic scenarios, i.e., the best result of the decision x. Therefore, the objective median $M(x, y)$ in uncertainty can represent the effect of some decisions. In the minimization optimization, median $M(x, y)$ should also be minimized to promote decision effectiveness. Similarly, the objective width $W(x, y)$ in uncertainty can represent the robustness of some decisions. In other words, a smaller $W(x, y)$ means more stable decision results.

However, a decision may be difficult to balance efficiency and robustness. As shown in Fig. 2, the median and width of x_1 are inferior to the other decisions x_2 and x_3, thus the performance of decision x_1 is obviously worse. Nevertheless, there may be no complete advantages between multiple decisions, e.g., x_2 and x_3, $M_2 > M_3$, $W_2 < W_3$. In this context, the decision result will depend on the subjective preference of the decision-maker.

The median and width are established based on the objective boundary of the pessimistic and optimistic values, i.e., $\max_y[F(x, y)]$ and $\min_y[F(x, y)]$, and these extreme values tend to be obtained at the boundary of the interval variable y. The existing interval optimization defaults that scenario interval values have the same probability of occurrence, i.e., uniform distribution. In fact, the scenario value y usually follows a distribution with significant probability difference [32], although it is difficult to obtain. In this distribution, if the pessimistic and optimistic scenario of a decision is in the low-probability boundary value, the median and width in the characterization of efficiency and robustness may be not accurate enough, because these extreme scenarios are unlikely to occur and these objective values may be overestimated or underestimated in practice.

To optimize the selection of pessimistic and optimistic scenarios, this article formulates a standard deviation degree $D(y)$ as a probability correction. Mathematical formulas can be expressed as (21).

$$D(y) = |y - \bar{y}|_{\mathrm{d}}, \tag{21}$$

where $|\cdot|_{\mathrm{d}}$ is defined as the cumulative norm of vector elements, and $\bar{y}$ denotes the scenario with maximum probability, i.e., frequent scenario, which is easy to obtain even in the condition of multiple scattered areas. $D(y)$ is the *Euclidean distance* between the frequent scenario $\bar{y}$ and pessimistic or optimistic scenarios y. Generally, the shorter *Euclidean distance* $D(y)$ means more probability. Although this simple distance cannot be used to describe the exact distribution, it is enough to be a universal correction degree to promote the probability of pessimistic or optimistic scenarios. Take Gaussian distribution as an example, as shown in Fig. 3, when the optional interval is reduced from the ideal boundary to the optimal boundary, the selected scenario will have a higher probability. Here, the distribution means is the frequent scenario, and $D(y)$ represents interval deviation.

However, if the selected optimal boundary is too narrow, the optimization result is hard to adapt to the extreme conditions in the whole ideal boundary. Therefore, this decision process also needs trade-offs and compromises. A modified interval optimization model is proposed in (22).

$$\begin{aligned} \min_x \quad & [M(x,y),\ W(x,y),\ D(y)] \\ \text{s.t.} \quad & h(x,y) = 0,\ g(x,y) \le 0 \\ & y \in [y^{\mathrm{L}}, y^{\mathrm{U}}], \end{aligned} \tag{22}$$

where $h(x, y)$ and $g(x, y)$ are equality and inequality constraints, while y^{L} and y^{U} are lower and upper bounds of interval variables. This section introduces the IO idea and the correction method considering probability. For the IO model as (22), we note the non-convex optimization process of the min max problem. The next section will introduce the corresponding equivalence and solution method.

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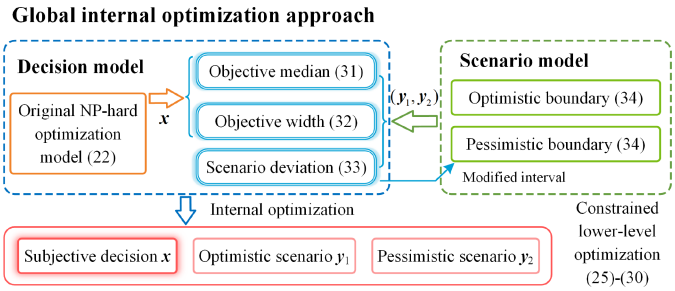


Fig. 4. Global interval optimization approach.

IV. Global Interval Optimization Approach in High-Order and Non-Convex Model

This article develops a global interval optimization approach to solve the above uncertain decision model and optimize the global optimal solution. In the high-order and non-convex original optimization as (18), the median and width of the original objective are established based a process of scenario selection. Specifically, the optimization of $M(x,y)$ and $W(x,y)$ can be formulated as (23) and (24).

$$\min_x M(x,y) = \left\{ \min_x \max_y [F(x,y)] + \min_x \min_y [F(x,y)] \right\} \Big/ 2, \tag{23}$$

$$\min_x W(x,y) = \left\{ \min_x \max_y [F(x,y)] - \min_x \min_y [F(x,y)] \right\} \Big/ 2, \tag{24}$$

where $\min_x \max_y$ is a bi-level optimization, which means that the subjective decision x is optimized based on the scenario selection result involving y. In Fig. 4, the GIO approach divides this bi-level relationship into a decision model and a scenario model. Here, the decision model is used to optimize efficient and robust decisions, i.e., $\min_x$, while the scenario model is used to select pessimistic and optimistic scenarios to generate objective median and width, i.e., $\max_y$ and $\min_y$. In addition, this approach formulates a constrained scenario model to achieve the integration of the subjective decision and scenario selection.

In fact, if the scenario problem can be equivalent to the constraints, IO can be established as a complete model, although its original problem is high-order and non-convex. This constrained derivation is the core of establishing the complete optimization model of the GIO. We firstly formulate one of the lower optimization $\min_y[-F(x,y)]$ or $\min_y[F(x,y)]$ as *Lagrange* function. In (25), '$\pm$' is used to present the above two situations, comprehensively.

$$\mathscr{L} = \pm F(x,y) + \upsilon^\top \cdot h(x,y) + \lambda^\top \cdot g(x,y), \tag{25}$$

where υ and λ are dual variables in constraints involving y. On this basis, the corresponding sufficient condition for applying the GIO approach is present as follows.

Sufficient condition: For $\theta \in [0,1]$ and $\forall x_0$, if $\mathscr{L}(x,y)|_{x=x_0}$ follows:

$$\begin{aligned} &\mathscr{L}(x_0, \theta \cdot y_1 + (1-\theta) \cdot y_2) \\ &\quad = \theta \cdot \mathscr{L}(x_0, y_1) + (1-\theta) \cdot \mathscr{L}(x_0, y_2), \end{aligned} \tag{26}$$

there will be no gap between the *Lagrange* dual optimal $\mathscr{L}^*$ and the original optimal F^*, i.e., $\mathcal{L}^* = F^*$.

Proof: Set A is defined as $A = \{(m,n,k) | \exists y \in \text{int } D,\ g(x_0, y) \le m,\ h(x_0,y) = n, F(x_0,y) \le k\}$. If $\mathscr{L}(x,y)|_{x=x_0}$ follows (26), $g(x_0,y)$, $h(x_0,y)$ and $F(x_0,y)$ are linear or affine for unique variable y, and set A is convex. In addition, another convex set B is defined as $B = \{(0,0,t) | t < F^*\}$, where A and B are separate. The reason is that $F^*(x_0,y) \le t < F^*$ conflicts with the F^* optimal definition in $A = \{(0,0,t)\}$.

For two separate sets, it is existing a hyperplane that is non-intersect with both, i.e., separating hyperplane $\bar{\lambda}^\top \cdot x + \bar{\nu}^\top \cdot y + \mu \cdot z = 0$. Furthermore, (27) is established in the set A and B, and γ is a certain constant, $\gamma \ge 0$.

$$\bar{\lambda}^\top \cdot m + \bar{\nu}^\top \cdot n + \mu \cdot k \ge \gamma,\ \bar{\lambda} \cdot 0 + \bar{\nu} \cdot 0 + \mu \cdot t \le \gamma, \tag{27}$$

where $\bar{\lambda} \succ 0$ and $\mu \ge 0$, otherwise the lower bound condition will not hold in (27). Accordingly, for $\forall t \le F^*$, $\mu \cdot t \le \gamma$ and

$$\mu \cdot F(x_0,y) + \bar{\upsilon}^\top \cdot h(x_0,y) + \bar{\lambda}^\top \cdot g(x_0,y) \ge \gamma \ge \mu \cdot F^*, \tag{28}$$

i.e., $\min_y \mathcal{L}(x_0,y)|_{\lambda=\bar{\lambda}/\mu, \nu=\bar{\nu}/\mu} \ge F^*$. On the other hand, due to $g(x_0,y) \le 0$, and $\lambda \ge 0$, $\min_y \mathcal{L}(x_0,y) \le F^*$. Therefore, $\min_y \mathcal{L}(x_0,y) = d^* = F^*$, and there will be no gap between the *Lagrange* dual optimal $\mathscr{L}^*$ and the original optimal F^*, i.e., $\mathcal{L}^* = F^*$. ■

Based on the sufficient condition, we could formulate the optimality constraints (29) and (30) of lower optimization according to the *Karush-Kuhn-Tucker* conditions [33].

$$\nabla_y \mathscr{L}_y = 0,\quad \lambda \circ g(x,y) = 0, \tag{29}$$

$$h(x,y) = 0,\quad g(x,y) \le 0,\quad \lambda \ge 0, \tag{30}$$

where $\circ$ denote *Hadamard* product, i.e., multiplication of corresponding vector elements. Based on (29) and (30), the scenario selection of $\max_y[-F(x,y)]$ or $\max_y[F(x,y)]$ is equivalently constrained. This is an effective approach to simplify the lower optimization in the bi-level. In the above derivation process, x is parameterized, i.e., this process is not depending on the original problem.

According to the equivalent constraints, the HNIO could be further established as a single-level optimization. Firstly, y_1 and y_2 are defined as interval variables representing pessimistic and optimistic scenarios separately. In the different optimal constraints to y_1 and y_2, the corresponding median, width, and standard deviation degree are reformulated as (31)–(33), equivalently omitting the process of $\min\ \max$.

$$M(x,y_1,y_2) = [F(x,y_1) + F(x,y_2)]/2, \tag{31}$$

$$M(x,y_1,y_2) = [F(x,y_1) - F(x,y_2)]/2, \tag{32}$$

$$D(y_1,y_2) = |y_i - \bar{y}_i|_d,\ i = 1,2. \tag{33}$$

In order to limit these scenario values, we formulate the optimality constraints of lower optimization in the corresponding dual variables $(\upsilon_1, \upsilon_2, \lambda_1, \lambda_2)$ and *Lagrange* function $(\mathscr{L}_1, \mathscr{L}_2)$. According to the constraints of (25)–(33), the complete GIO model considering probability correction could be established

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TABLE I
SIMULATION CASE PARAMETER

Case	Optimization approach	Distribution	Mean	Std	Interval boundary
Case 1	Stochastic optimization	*Gaussian*	Frequent scenario	$[\mathrm{Abs}\cdot \mathrm{Mean}/3]\%$	—
Case 2	Robust optimization	—	—	—	$\pm 10\%$
Case 3	GIO	*Uniform*	Frequent scenario	—	$\pm 10\%$
Case 4	GIO with correction	—	Frequent scenario	—	$\pm 10\%$

as (34).

$$\begin{aligned}
\min_{x,y_1,y_2} \quad & [M(x,y_1,y_2),\ W(x,y_1,y_2),\ D(y_1,y_2)] \\
\text{s.t.} \quad & h(x,y_i)=0,\ g(x,y_i)\le 0 \\
& \mathscr{L}_1 = -F(x,y_1)+\upsilon_1^{\top}\cdot h(x,y_1)+\lambda_1^{\top}\cdot g(x,y_1) \\
& \mathscr{L}_2 = F(x,y_2)+\upsilon_2^{\top}\cdot h(x,y_2)+\lambda_2^{\top}\cdot g(x,y_2) \\
& \nabla_{y_i}\mathscr{L}_i=0,\ \lambda_i\circ g(x,y_i)=0 \\
& y_i\in[y_i^{\mathrm{L}},y_i^{\mathrm{U}}],\ \lambda_i\ge 0,\ i=1,2. \qquad (34)
\end{aligned}$$

The complete GIO model is the explicit three-objective problem. In this article, the Augmented ϵ-constraint method is applied to obtain the three-objective *Pareto* solution [34], as shown in Appendix A. The GIO approach does not depend on the nature of the original optimization problem, and independently deduces the equivalent constraints of scenarios selection involving interval variable y. Therefore, this approach is able to be applied to the non-convex optimization problem of multi-class power dispatching. The process of interval optimization is modeled as a complete solvable model, thus the corresponding convergence solution is theoretically globally optimal.

V. CASE STUDY

A. Case Setup

This section is devoted to verifying the superiority of the proposed global interval optimization approach and the effectiveness of the DCCHSS collaborative response. Correspondingly, Table I sets four cases to compare the optimization performance of different approaches in uncertain WP conditions.

In Case 1, stochastic optimization (SO) based on the probability scenario is applied as a comparison to verify the superiority of the GIO approach, where the wind speed distribution is assumed to follow the common *Gaussian* distribution [32]. The mean and standard deviation (Std) in these distributions separately are the values of the most frequent scenarios and $[\mathrm{Abs}\cdot \mathrm{Mean}/3]\%$, where Abs is the absolute value of the interval range. This Std means that the supplied WP interval is limited to $\pm 10\%$. Case 2 represents the solution performance of robust optimization (RO) [35]. In this case, the uncertainty variables follow the box set, and a budget uncertainty set is introduced to improve the consevativeness. The specific robust model is visible in Appendix B. Ulteriorly, the traditional IO model is set as Case 3 to verify the effective improvement by considering probability correction in Case 4. Furthermore, in order to quantify the decision effect of the above cases in the specific uncertain scenarios, this article tests these decisions performance in the 1000 random scenarios of wind speed. The generation of scenarios is based on *Latin hypercube sampling approach* [36], and the speed-power model of wind turbine is provided in Appendix C.

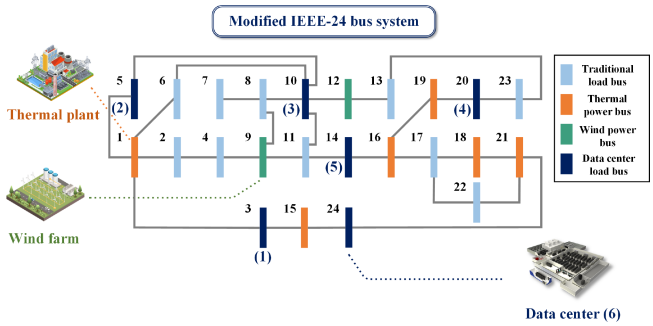


Fig. 5. Modified IEEE-24 bus system.

TABLE II
DATA CENTER PARAMETER [30]

PUE	SV number ($X_{i,t}^{sv}$) /10^5	Frequency (f) /GHz	Static power (P_{st}^{sv}) /W	Dynamic factor (k) W/GHz	Processing time (D) /s
1.2	[1.8,1.8,2.2,2, 2.2,2.3]	0.8∼3.0	53	6.5	0.5

TABLE III
HYDROGEN STORAGE SYSTEM PARAMETER [17], [37]

Efficiency (P2H) (η^{el})	Density (P2H) (ρ)	Efficiency (H2P) (η^{f})	Density (H2P) (μ)	H_2 storage efficiency ($\eta^{\mathrm{ch}}/\eta^{\mathrm{dis}}$)	Storage capacity ($Q_{\max}$)
0.75	0.20	0.60	0.295	0.95	[2.25,2.25,2.70, 1.80, 2.70]

The above case simulations are based on the modified IEEE 24-bus system [38], as shown in Fig. 5. Two wind farms including 200 wind turbines are connected to buses 9 and 12, whose supplied power will be shown later. In addition, 6 distributed data centers and corresponding HSS systems are placed at buses [3, 5, 10, 14, 20, 24]. The specific parameters of data centers and HSS are shown in Tables II and III, and these value are partially modified to adapt the simulation cases.

B. Analysis of Decision Performance

The three-objective *Pareto* solution set of the GIO approach is shown in Fig. 6. The objectives of median, width and deviation respectively characterize the effectiveness, robustness, and degree of probability correction. Here, each curve represents a unique *Pareto* solution, and the objective value corresponds to the position of the intersection coordinate axis. The *Pareto* set has many optional solutions and all the solution curves intersect. It means each solution has no complete advantage: a solution

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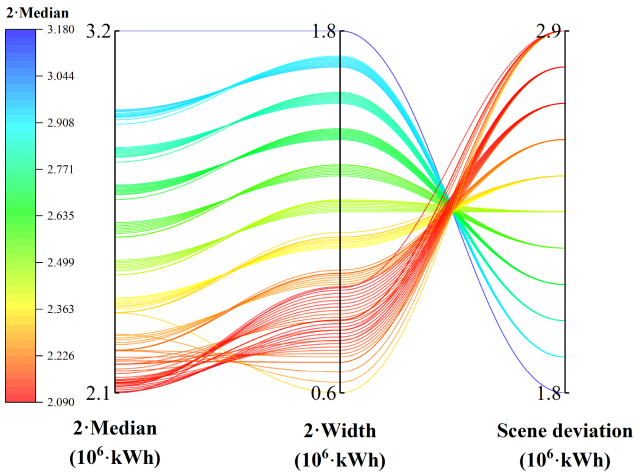


Fig. 6. Three-objective Pareto of GIO.

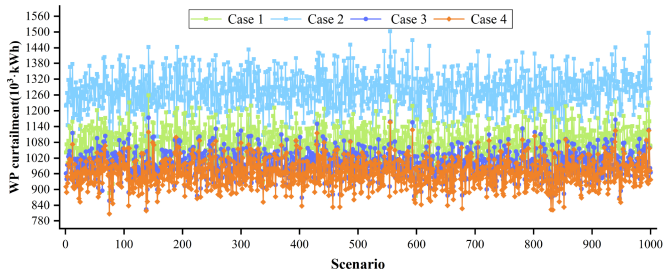


Fig. 7. Performance comparison of efficient solution in four cases.

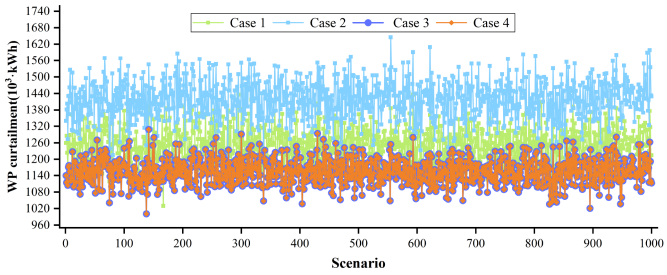


Fig. 8. Performance comparison of robust solution in four cases.

performing better in one objective will perform worse in another objective.

To compare decision performance in the same conditions, we test all the decisions in four cases in 1000 random scenarios and quantify the corresponding WP reduction. On this basis, the best decision (Optimal effectiveness or robustness) in each case is selected to compare. Case 2 provides two robust solutions. One is the robust solution considering the budget uncertainty set, and the other one is the most conservative robust solution. The test results are shown in Figs. 7 and 8, where the abscissa denotes the serial number of the random scenarios, and the ordinate denotes the testing performance of WP curtailment for the decision in four cases.

In the scenario test of the optimal efficient solution, the distribution of WP curtailment in Case 4 is obviously the lowest.

TABLE IV
SPECIFIC DECISION PERFORMANCE

	Efficient solution				Robust solution			
	Mean	Median	Std	Width	Mean	Median	Std	Width
Case 1	1.080	1.067	0.058	0.192	1.255	1.233	0.056	0.203
Case 2	1.278	1.278	0.052	0.224	1.416	1.400	0.050	0.245
Case 3	0.999	0.999	**0.050**	0.175	**1.156**	**1.115**	**0.046**	**0.154**
Case 4	**0.963**	**0.982**	0.053	**0.174**	**1.156**	**1.115**	**0.046**	**0.154**

Specifically, the worst testing result caused by the efficient solution in Case 4 is 1156.63 $\times$ 10^3·kWh in Scenario 555, which has less curtailment than 1258.10 $\times$ 10^3·kWh in Case 1 (Scenario 142), 1502.46 $\times$ 10^3·kWh in Case 2 (Scenario 555) and 1173.85 $\times$ 10^3·kWh in Case 3 (Scenario 142). Similarly, the best testing result caused by efficient solution in Case 4 is 807.67 $\times$ 10^3·kWh in Scenario 75, which has less curtailment than 874.97 $\times$ 10^3·kWh in Case 1 (Scenario 599), 1053.82 $\times$ 10^3·kWh in Case 2 (Scenario 75) and 824.59 $\times$ 10^3·kWh in Case 3 (Scenario 138). In the scenario test of the optimal robust solution, the decisions in Case 3 and Case 4 have the same effect, which means that the probability correction will not adversely affect the robustness compared with the original IO model. Moreover, the worst testing result caused by the efficient solution in Case 4 is 1308.56 $\times$ 10^3·kWh in Scenario 142, which has less curtailment than 1435.70 $\times$ 10^3·kWh in Case 1 (Scenario 295) and 1645.27 $\times$ 10^3·kWh in Case 2 (Scenario 555). The best testing result caused by the efficient solution in Case 3 is 1000.93 $\times$ 10^3·kWh in Scenario 138, which have less curtailment than 1029.57 $\times$ 10^3·kWh in Case 1 (Scenario 167) and 1155.40 $\times$ 10^3·kWh in Case 2 (Scenario 830). Thus, by considering probability correction in GIO, we can find that the WP dispatch achieves the more advantageous boundary of power curtailment in uncertain conditions.

Table IV provides the comparison of specific numerical indicators among the four cases, and the formulas of these indicators are provided in Appendix C. Based on the important indicators (Mean, Median, Std, Width), the efficient solution in Case 4 reduces the mean by 3.7%, median by 1.7% and width by 0.5% in scenario testing compared with Case 3. For Case 1, the desceases are the mean by 12.1%, median by 8.7%, std by 9.4%, and width by 10.3%. For Case 2, the desceases are the mean by 32.7%, median by 30.1%, and width by 28.7%. Correspondingly, the robust solutions of Case 4 and Case 3 have the same performance. Additionally, robust solution in Case 4 reduces the mean by 8.7%, median by 10.6%, std by 21.7%, and width 31.8% in scenario testing compared with Case 1. Contrast to the conservative Case 2, the robust solution has the more decreases of 22.5% in mean, 14.7% in median, 8.7% in std and 59.1% in width. These comparisons illustrate that efficient and robust solutions present more outstanding decision performance.

The above simulation result proves that the proposed GIO approach considering probability correction has significant superiority for the stochastic optimization, robust optimization and IO. The reason is that the global optimal solutions in the GIO mean the best performance theoretically and these solutions involving probability consideration are more suitable for the

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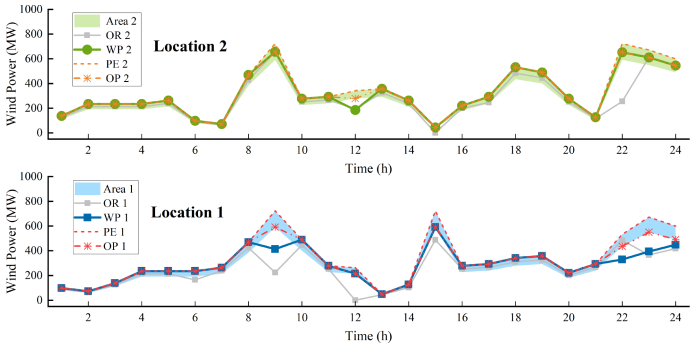


Fig. 9. WP-connection strategy and curtailment estimate of efficient solution.

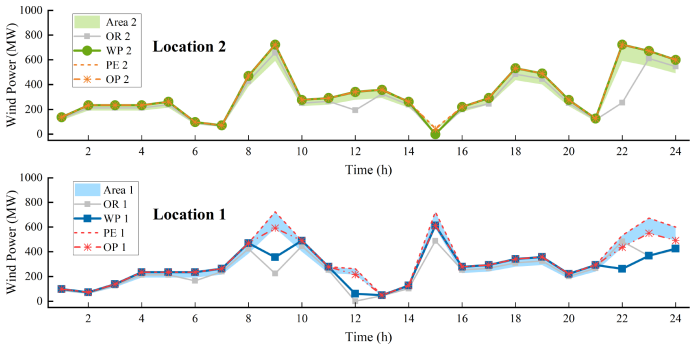


Fig. 10. WP-connection strategy and curtailment estimate of robust solution.

actual differential distribution. The specific case decisions of wind connection and the actual wind supply curve are provided in Appendix D, including total WP prediction, actual available WP, WP connection decision, etc. In the next subsection, the specific information and GIO decisions of each wind farms are visible. Based on these, we would further analyze the decision impacts, as well as the characteristcs of optimistic and pessimistic scenarios.

C. Analysis of WP Connection in the Efficient and Robust Solution

For the efficient and robust solution, as shown in Figs. 9 and 10, this section specifically will analyze the WP connection (WP) strategies in the two solutions. Herein, the blue and green areas represent the interval of applied WP, and the pessimistic and optimistic scenarios (PE and OP) are the upper bound of the WP connection in the decision. Therefore, the selection of these scenarios will influence the actual WP utilization. In fact, the decision performance of both solutions is excellent. For example, at 1:00–8:00 and 16:00–21:00, the maximum possible supplied WP is fully utilized in all wind farms. The optimized connection power is higher than the original (OR) WP demand. For example, at 9:00, 12:00 and 22:00, the connected WP at locations 1 and 2 increased significantly, which indicates that the anti-peak characteristic of these periods has been alleviated.

Therefore, the performance difference of the solution depends on the selection of a special slot. The efficient solution tends to connect the WP of two locations in the well-distributed, such as 214.76×10^3·kWh in location 1 at 12:00 and 185.69×10^3·kWh in location 2. Similarly, the same is true at 22:00–24:00. This trend causes a significant reduction of WP curtailment compared with the estimated value, when supplied WP is not high-level in both wind farms. The robust solution tends to connect to WP in the location had more supplied capability. This preference is obvious at 12:00 and 22:00–24:00. Take 12:00 as an example in Fig. 10, location 2 with more supply has more WP connection and location 1 has less. With the WP supply fluctuation, there will be no curtailment in location 2 and a slight fluctuation of curtailment in location 1. Therefore, the fluctuation width and variance of WP curtailment are relatively lower in robust solutions. Furthermore, by comparing the WP interval with the pessimistic and optimistic scenarios, we can verify the conclusion in the introduction, i.e., these extreme scenarios of the high-order non-convex problem usually do not exist on the interval boundary.

D. Analysis of Specific DCCHSS Collaborative Response

In this subsection, efficient solutions and corresponding responses are selected to analyze. Fig. 11 shows the specific response of the six data center loads and configured HSS, which includes the IT data workload and the corresponding increase (IT IN) and decrease (IT DE) workload, the discharging (DISCH) and charging (CH) power of hydrogen storage, the regular load (RG) and the total load (TO). Here, the data center 1∼4 increase the power demand by moving in the data workload and increasing the server working frequency. Accordingly, the data center 5∼6 reduce the power demand to release the excessive load in corresponding transmission branches by the emigration of data workload, which achieves more utilization of wind power. It is worth noting that the total power load of multiple data centers is increasing during the wind power excess period, even if the total amount of data workload is constant. It is because the data centers aim to adopt a higher working frequency to improve their power demand in dispatching response.

Moreover, during the peak periods of WP supply, such as 9:00, 15:00 and 22:00, the configured hydrogen storage systems electrolyze hydrogen to increase power demand and utilize excess WP. During the valley periods of WP supply, such as 2:00, 6:00 and 21:00, the hydrogen storage systems apply hydrogen fuel cells to supply power to the corresponding data center, which is beneficial to reduce external power demand for data centers. In this simulation, HSS is more inclined to charge and store hydrogen. Because it is difficult to fully utilize the excess WP of the regional power system in some periods, and the energy storage power supply as an alternative is more reasonable in the data center operation.

E. Economic Analysis of Collaborative Response

In order to quantify the economic impacts of DCCHSS, the profit function, difference between the original and the responding, can be formulated as (35). For data center operator, it consists of the commendation of WP absorption from the main electricity company, the migration fee of data workload, and the

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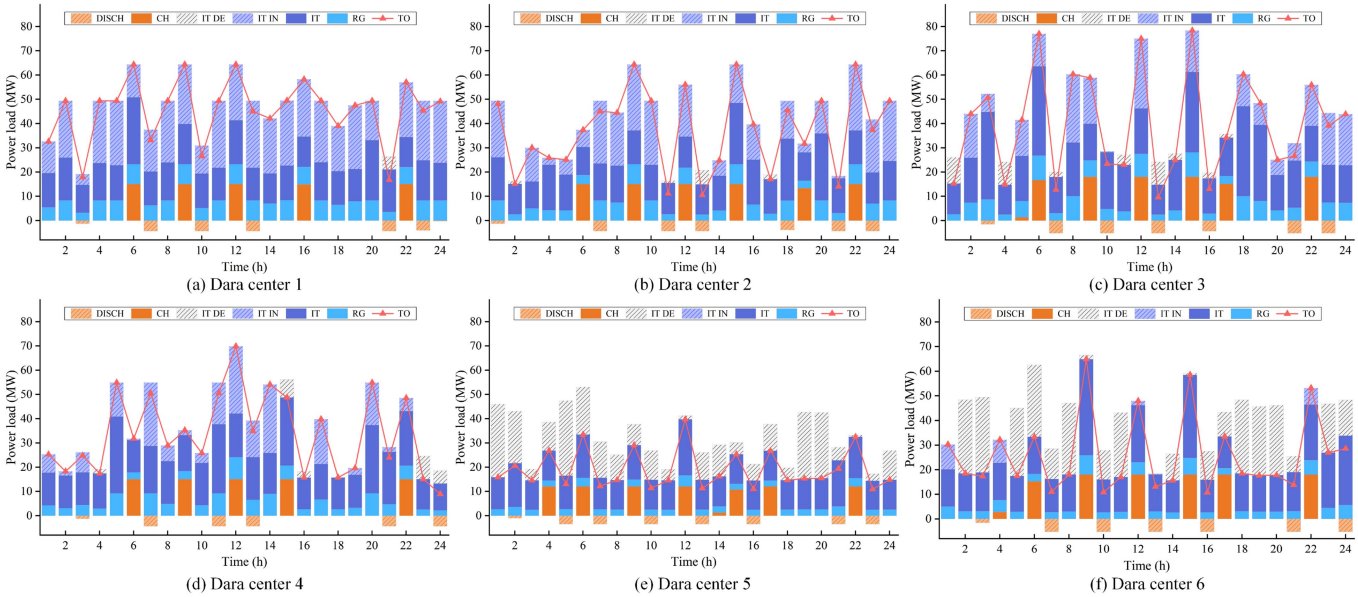


Fig. 11. Specific DCCHSS collaborative response in the multiple data centers. (a) Dara center 1. (b) Dara center 2. (c) Dara center 3. (d) Dara center 4. (e) Dara center 5. (f) Dara center 6.

loss of storage systems.

$$\text{profit} = \alpha \cdot \sum_{w=1}^{\mathrm{W}} \sum_{t=1}^{\mathrm{T}} (P_{w,t}^{\mathrm{WP}} - P_{w,t}^{\mathrm{WP,or}}) \cdot \Delta t - \beta \cdot \sum_{t=1}^{\mathrm{T}} \lambda_{i,t}^{\mathrm{MI}} - \gamma \cdot \sum_{i=1}^{\mathrm{N}} \sum_{t=1}^{\mathrm{T}} P_{i,t}^{\mathrm{loss}} \cdot \Delta t \tag{35}$$

where α is financial commendation for increased WP utilization, 0.15·\$/kWh; β represents the transmission fee, 0.1·\$/GB according to Microsoft Azure; γ denotes the constant electricity price, 0.2·\$/kWh. The difference of actual wind power $P_{w,t}^{\mathrm{WP}}$ and the original wind power $P_{w,t}^{\mathrm{WP,or}}$ is increased value. $P_{i,t}^{\mathrm{loss}}$ is the loss power of storage system, $P_{i,t}^{\mathrm{loss}} = (1 - \eta^{\mathrm{el}}\eta^{\mathrm{f}}\eta^{\mathrm{ch}}\eta^{\mathrm{dis}}) \cdot P_{i,t}^{\mathrm{Ch}}$ for HSS.

Comparing with the battery storage system (BSS), HSS generally has the more storage capacity. Thus, HSS is able to supply power to data center in the long-term. However, the advantage of BSS is the high-level conversion efficiency. In this context, the quantized economic performances are also analyzed in this subsection. Instead of HSS, we formulate the BSS model in the same case of data centers for comparision. Specifically, the BSS conversion model is presented as (36).

$$E_{i,t}^{\mathrm{BSS}} = (1 - r_{sd}) \cdot E_{i,t-1}^{\mathrm{BSS}} + \eta^{\mathrm{I}} \cdot P_{i,t}^{\mathrm{B,ch}} - P_{i,t}^{\mathrm{B,dis}} / (\eta^{\mathrm{I}} \cdot \eta^{\mathrm{B}}), \tag{36}$$

where $E_{i,t}^{\mathrm{BSS}}$ represents the battery energy stored; $P_{i,t}^{\mathrm{B,ch}}$ and $P_{i,t}^{\mathrm{B,dis}}$ are charging power and discharging power; r_{sd} denotes the attenuation coefficient of battery, 0.004%; η^{I} and η^{B} represent the inverter efficiency and battery efficiency, 97% and 95% [11]; The maximum of $E_{i,t}^{\mathrm{BSS}}$ is set as [3.5, 4.5, 5.0, 4.5, 4.0, 5.0] MW · h. The charging power and discharging power follow the boundary constraints as (37)–(38).

$$P_{i,t}^{\mathrm{B,ch,min}} \le P_{i,t}^{\mathrm{B,ch}} \le P_{i,t}^{\mathrm{B,ch,max}}, \tag{37}$$

$$P_{i,t}^{\mathrm{B,dis,min}} \le P_{i,t}^{\mathrm{B,dis}} \le P_{i,t}^{\mathrm{B,dis,max}}, \tag{38}$$

The discharging power is not grid-connected, therefore the discharge power should be not higher than the power demand of data centers. Additionally, the charging and discharging processes cannot exist at the same time. These constraints are established as (39).

$$P_{i,t}^{\mathrm{B,dis}} \le P_{i,t}^{\mathrm{DC}}, \quad P_{i,t}^{\mathrm{B,ch}} \cdot P_{i,t}^{\mathrm{B,dis}} = 0. \tag{39}$$

On this basis, the comparison of case setting is visible in Table V. For controlling the testing difference, Cases 5 and 6 are simulated in the same certain wind speed scenario. Here, the adjustment of IT server status and workload migration are available in the two cases. Case 5 deploys large-scale hydrogen storage as the above cases, yet Case 6 deploys the battery storage system.

TABLE V
CASE SETTING OF COMPARISION

Case	Adjustment of server status	Workload migration	Hydrogen storage	Battery storage
Case 5	✓	✓	✓	
Case 6	✓	✓		✓

The specific performances of the two cases are presented in Table VI. It is clarify that the collaborative response of the data center and HSS is able to profit from WP absorption commendation, although relatively high energy loss causes the cost adding. Due to the higher conversion efficiency, Case 6 with BSS performs less energy loss (1.39×10^4· kWh) than Case 5 (2.95×10^5·kWh). It directly causes more responding

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TABLE VI
PERFORMANCE OF CASE COMPARISON

Case	Increased wind energy /kWh	Storage energy loss /kWh	Data migration /GB	Total profit /$	Available storage /kWh
Case 5	$\mathbf{1.41\times10^6}$	2.95×10^5	$\mathbf{3.81\times10^5}$	1.14×10^5	$\mathbf{1.53\times10^5}$
Case 6	1.12×10^6	$\mathbf{1.39\times10^4}$	3.96×10^5	$\mathbf{1.26\times10^5}$	5.90×10^3

profit in Case 6 (1.26×10^5\$) than in Case 5 (1.14×10^5\$). From the perspective of responding ability, the data center load in Case 5 absorbs more WP (1.41×10^6·kWh) than in Case 6 (1.12×10^6·kWh). Meanwhile, Case 5 needs less migration (3.81×10^3·GB) than Case 6 (3.96×10^3·GB). It is most important that the significant superiority in the available storage (loss considered), 1.53×10^5·kWh in Case 5 and 5.90×10^3·kWh in Case 6. It means that HSS has longer continuing to support power, which is the base requirement of energy system. In fact, the electricity price exists the obvious difference in each period. The considerable energy storage of HSS enables the data center to avoid electricity purchases in the peak period of electricity prices. This strategy could contribute to covering the profit difference compared with BSS.

VI. CONCLUSION

This article proposes the collaborative response of the data center coupled with hydrogen storage system to boost wind power utilization. Considering uncertain wind power supply, the global interval optimization approach is developed to optimize decisions and improve the accuracy in the actual scenario test. The derivation of the GIO does not depend on the specific properties of the original problem. Therefore, it can be applied to the high-order non-convex optimization as the above collaborative response and achieve the global optimal solution. Accordingly, in the case study based on the IEEE 24-bus system, we summarize the following conclusions.

1) The collaborative response of the data center coupled with hydrogen energy system can effectively alleviate the wind power anti-peak characteristic. Compared with the original status, this response achieves a reduction in wind power curtailment.
2) The global interval optimization approach is obviously superior to the stochastic optimization, robust optimization, and traditional interval optimization approach. The mean of GIO solution performs the improvement of 12.1% and 32.7% compared with SO and RO in the efficient solution, especially 3.7% contrast to GIO without correction. The standard deviation of GIO solution performs the improvement of 21.7% contrast to SO in the robust solution, especially 8.7% compared with RO. This is because GIO achieves the global optimality and the introduction of probability correction enhances the decision performance and the adaptability in the actual uncertain scenarios.
3) It is potential for data center operators to obtain financial profit through collaborative response. Considering the commendation of WP absorption, extra income is able to not only cover the cost of HSS and migration fee, but also achieve the profit of 1.14×10^5\$.

In future research, we plan to consider the comprehensive utilization of multiple internal energy sources in the data center, such as hydrogen storage, battery storage, waste heat recovery, and diesel generator. It is expected to further improve the flexibility of the data center to respond to the power system demand.

APPENDIX A
AUGMENTED ϵ-CONSTRAINT ALGORITHM

Algorithm 1: Augmented ϵ-constraint Algorithm.

Input: Three objectives optimation $\{F_1, F_2, F_3\}$, specific small number ϵ, slack variable s_i, split coefficient G
Output: Pareto solution PF

Structure of payoff table P, $P \in \mathbb{R}^{3\times3}$:
$P_{i,i} = \min[F_i],\ (i = 1, 2, 3)$
$P_{i,I} = \min[F_I]$, s.t. $F_i = P_{i,i}, (I = i + 1, \textbf{if } I > 3, \textbf{then } I = 1)$
$P_{i,J} = \min[F_J]$, s.t. $F_i = P_{i,i}, F_I = P_{i,I}, (J = i + 2, \textbf{if } J > 3, \textbf{ then } J = J - 2)$
Division of pareto points:

$$e_i^g = \begin{cases} e_i^{g-1} + (P_{\max,i} - P_{i,i})/G, & g = 2, \ldots, G \\ P_{i,i}, & g = 1 \end{cases}$$

for $g_2 = 1, 2, \ldots, G$ **do**
  for $g_3 = 1, 2, \ldots, G$ **do**
    $\min \quad F = F_1 - \varepsilon \cdot [s_2/(P_{12} - P_{22}) + s_3/(P_{31} - P_{33})]$
    s.t. $F_2 + s_2 = e_2^{g_2},\ F_3 + s_3 = e_3^{g_3}$
    $\mathrm{PF}(g) = (F_1, F_2, F_3)$
  end for
end for

APPENDIX B
ROBUST COUNTERPART OF DCCHSS

In the Case 2 of Section V, the solution of robust optimization is applied to be comparison. In this section, the specific robust counterpart model is introduced [35]. Firstly, the original optimization problem can be formulated as (B.1). In this context, the uncertain variables are following box set $y \in [y^{\mathrm{L}}, y^{\mathrm{U}}]$, which is similar to the internal optimization.

$$\begin{aligned} \min_x \quad & F(x, y) = \sum_{w=1}^{\mathrm{W}} (P_{w,t}^{\mathrm{WSP}} - P_{w,t}^{\mathrm{WP}}) \\ \text{s.t.} \quad & h(x, y) = 0,\ g(x, y) \le 0 \\ & y \in [y^{\mathrm{L}}, y^{\mathrm{U}}], \end{aligned} \tag{B.1}$$

where $h(x, y)$ and $g(x, y)$ are the general formula of the equality and inequality constraints. In fact, uncertain variables y only involve in inequality constraints $g(x, y)$. On this basis, the base model is revised by the robust theory, and the robust model can

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be formulated as (B.2).

$$\begin{aligned} \min_{x} \quad & F(x,y) = ||y-x||_1 \\ \text{s.t.} \quad & h(x)=0, \ \sup_{y}[\,g(x,y)] \le 0 \\ & y \in [y^{\mathrm{L}}, y^{\mathrm{U}}], \end{aligned} \tag{B.2}$$

where $||\cdot||_1$ is the first-order vector norm; $\sup[\cdot]$ represents the supremum, which guarantees the robust solution is flexible in all the extreme conditions. The specific inequality form is $[\,g(x,y)]_{w,t} = [x-y]_{w,t} = P^{\mathrm{WP}}_{w,t} - P^{\mathrm{WSP}}_{w,t} \le 0$. The corresponding counterpart model can be further established as (B.3), which is based on the dual theory for $\sup[_y\, g(x,y)]$.

$$\begin{aligned} \min \quad & F(x,y,\omega) = ||y-x||_1 \\ \text{s.t.} \quad & h(x)=0, \ l(x) \le 0, \ y \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \\ & -y_{w,t} = y^{\mathrm{L}}_{w,t} \cdot {}^1\omega_{w,t} + y^{\mathrm{U}}_{w,t} \cdot {}^2\omega_{w,t} + {}^3\omega_{w,t} \cdot x_{w,t} \\ & (1+{}^3\omega_{w,t}) \cdot x_{w,t} + y^{\mathrm{L}}_{w,t} \cdot {}^1\omega_{w,t} + y^{\mathrm{U}}_{w,t} \cdot {}^2\omega_{w,t} \le 0 \\ & {}^1\omega_{w,t} + {}^2\omega_{w,t} + {}^3\omega_{w,t} = -1 \\ & {}^1\omega_{w,t}, {}^3\omega_{w,t} \le 0, \ \ {}^2\omega_{w,t} \ge 0. \end{aligned} \tag{B.3}$$

where ${}^1\omega_{w,t}$, ${}^2\omega_{w,t}$ and ${}^3\omega_{w,t}$ are dual variables related the constraints of lower-upper boundary $[y^{\mathrm{L}}, y^{\mathrm{U}}]$ and $g(x,y)$; $l(x)$ are the inequality constraints no involved y. (B.3) also means the most conservative model in this uncertain condition.

In order to reduce conservativeness, the budget uncertainty set, $U = \{z : |z_{w,t}| \le 1, ||z||_1 \le \Gamma\}$, is introduced into the above robust model [39]. Herein, parameter Γ is the conservative degree, $\Gamma \in [\,0, N^{u,\max}\,]$; $N^{u,\max}$ is the number of uncertain variables, $N^{u,\max} = \mathrm{W} \cdot \mathrm{T}$. According to Ref. [39], the inequality constraints in the counterpart model could be formulated as (B.4)–(B.5).

$${}^5\omega_{w,t}: \ \sup_{z}[x_{w,t} - y^{\mathrm{U}}_{w,t} + z_{w,t} \cdot \hat{y}_{w,t}] \le 0, \tag{B.4}$$

$${}^4\omega_{w,t}: \ 0 \le z_{w,t} \le 1 \ \ , {}^6\omega_{w,t}: \sum_{z_{w,t} \in z} z_{w,t} \le \Gamma. \tag{B.5}$$

where $\hat{y}_{w,t} = y^{\mathrm{U}}_{w,t} - y^{\mathrm{L}}_{w,t}$. Accordingly, the complete robust counterpart model is presented as (B.5), where z constraints (B.4) and (B.5) are transferred to the dual problem of ω, as presented as (B.6).

$$\begin{aligned} \min \quad & F(x,y,z,\omega) = ||y-x||_1 \\ \text{s.t.} \quad & h(x)=0, \ \sum_{z_{w,t} \in z} z_{w,t} \le \Gamma \ , z \in [0,1] \\ & l(x) \le 0, \ \ x_{w,t} - y_{w,t} \le 0 \\ & y_{w,t} = y^{\mathrm{U}}_{w,t} - z_{w,t} \cdot \hat{y}_{w,t} \\ & \hat{y}_{w,t} \cdot z_{w,t} = {}^4\omega_{w,t} + (y^{\mathrm{U}}_{w,t} - x_{w,t}) \cdot {}^5\omega_{w,t} + \Gamma \cdot {}^6\omega_{w,t} \\ & {}^4\omega_{w,t} + \hat{y}_{w,t} \cdot {}^5\omega_{w,t} + {}^6\omega_{w,t} \ge \hat{y}_{w,t} \\ & {}^4\omega_{w,t}, {}^5\omega_{w,t} \ge 0, \ \ {}^6\omega_{w,t} = 0. \end{aligned} \tag{B.6}$$

In (B.6), more value of Γ means more conservative, because $y_{w,t}$ is closer to the lower boundary. In the case simulation, we select a moderate value, $\Gamma = N^{u,\max}/2$.

Appendix C
Supplemental Explanation of Case Setup

Part 1: Speed-power model of wind turbine The function of wind speed-power is considered piecewise [32]. The sampling wind speed is transfered to wind power by the (C.1).

$$P^{\mathrm{WSP}}_{w,t} = \begin{cases} 0 & 0 \le v < v_{ci} \\ a + bv^3 & v_{ci} \le v < v_{ra} \\ P_{ra} & v_{ra} \le v \le v_{co} \\ 0 & v > v_{co} \end{cases}, \tag{C.1}$$

where v denotes wind speed; v_{ci}, v_{ra} and v_{co} are the cut-in, rated and cut-out wind speeds. P_{ra} represents the rated power of wind turbine; a and b are the constant coefficient. In these sample scenarios, the wind speeds are set in the bound of cut-out wind speed.

Part 2: The follows are the formulas of the four indicators applied in Section V-B.

(*i*) Mean: In the testing scenarios, the mean of WP curtailment represent the average level of decision performance.

$$\mathrm{Mean} = \sum_{s=1}^{\mathrm{S}} P^{\mathrm{Cur}}_{s,total} \cdot \Delta t/\mathrm{S}, \tag{C.2}$$

where $P^{\mathrm{Cur}}_{s,total}$ denotes the total WP curtailment of decision in sample s.

(*ii*) Median: In the testing scenarios, the median of WP curtailment represents the middle level of decision performance.

$$\mathrm{Median} = 0.5 \cdot \left[\ \max(P^{\mathrm{Cur}}_{s,total}) + \min(P^{\mathrm{Cur}}_{s,total}) \ \right] \cdot \Delta t, \tag{C.3}$$

(*iii*) Standard deviation (Std): In the testing scenarios, the Std of WP curtailment represents the disperse degree of the decision performance.

$$\mathrm{Std} = \sqrt{\frac{1}{(\mathrm{S}-1)} \sum\nolimits_{s=1}^{\mathrm{S}} \left(P^{\mathrm{Cur}}_{s,total} \cdot \Delta t - \mathrm{Mean}\right)^2} \cdot \frac{1}{\mathrm{Mean}}, \tag{C.4}$$

(*iv*) Width: In the testing scenarios, the Std of WP curtailment represents the deviation range of the decision performance.

$$\mathrm{Width} = 0.5 \cdot \left[\ \max(P^{\mathrm{Cur}}_{s,total}) - \min(P^{\mathrm{Cur}}_{s,total}) \ \right] \cdot \Delta t, \tag{C.5}$$

In this context, the indicators of Mean and Median are mainly used to evaluate the efficiency of the decision, and the indicators of Std and Width are used to evaluate the robustness of the decision.

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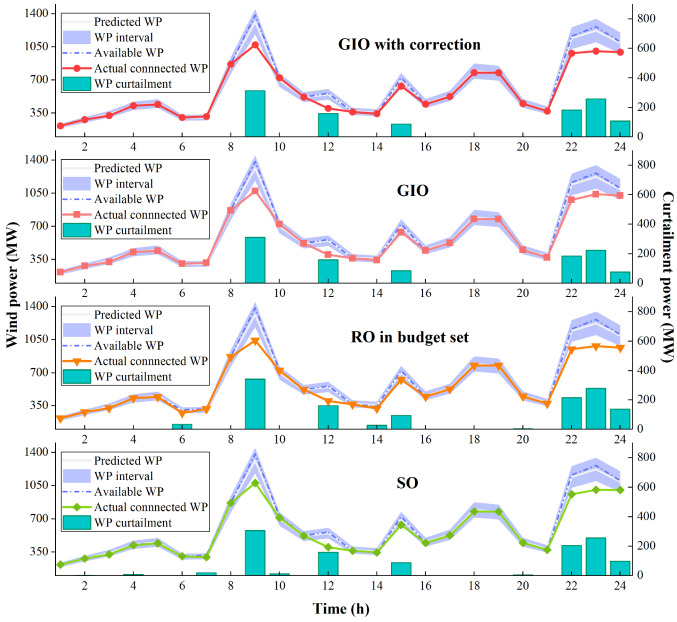


Fig. 12. Wind power information under efficient solutions of four cases.

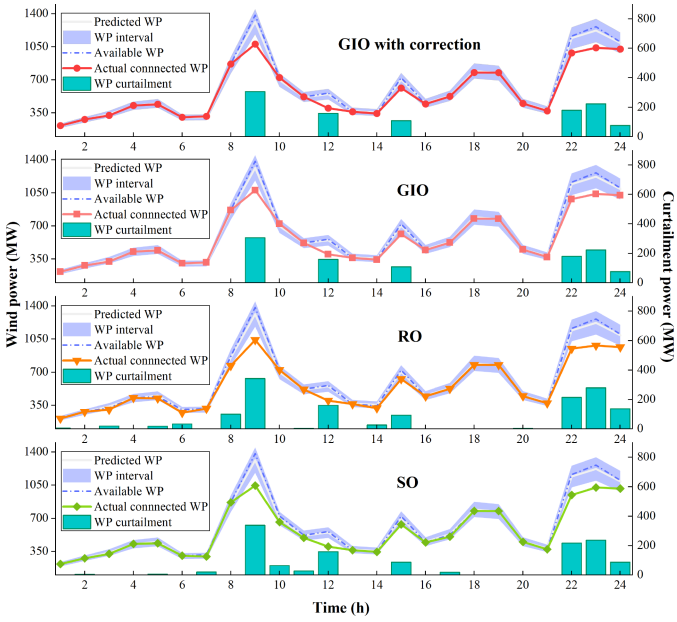


Fig. 13. Wind power information under robust solutions of four cases.

Appendix D
Specific Wind Power Information of Four Cases in a Single Day

The detailed wind power information for the four testing cases is depicted in Figs. 12 and 13. This information includes the predicted wind power value, fluctuation interval, actual available power, and the decision on connected power for all wind farms during a single day. The difference between the available and connected power represents the power curtailment value. In this context, the actual available power signifies the deviation from the standard predicted value within the fluctuation interval.

Focusing on the decisions regarding connected wind power, the fundamental trends are similar, aligning with the peaks and valleys of potential WP supply. These trends illustrate that all the applied methods for uncertainty management are effective to a certain extent. The numerical differences become more apparent in Table IV, although they might not be particularly conspicuous in the following figures. The primary reason for this lies in the fact that the configured flexible loads constitute only a relatively small portion of the total power consumption in the experimental power system, reflecting the real system scenario. Nevertheless, we can still discern the advantaamong the four decision methods mentioned above by comparing WP curtailment in each period.

Specifically, in the comparison of efficient solutions, the case of GIO considered correction performs 1102.71 MW of total power curtailment, which is more than 1034.76 MW of GIO without correction, and their curtailment periods are similar, i.e., 9:00, 12:00, 15:00, and 22:00–24:00. It is rational despite the performance results in Fig. 8 verifies the superiority of probability correction, it may be no adapt to all specific scenarios. The power curtailment situation of RO and SO show worse performances, 1286.38 MW and 1154.10 MW in values. The additional curtailment periods are 6:00, 14:00, and 20:00 for RO, as well as 2:00, 4:00, 7:00, 10:00, and 20:00 for SO. In the comparison of robust solutions, whether or not correction is considered, there is a similar solution for case 3 and case 4. The total power curtailment is 1048.96 MW, and the curtailment periods are also 9:00, 12:00, 15:00, and 22:00–24:00. The power curtailment situation of RO and SO show worse performance, 1434.05 MW and 1262.81 MW in values. The additional curtailment periods are 1:00, 3:00, 5:00–6:00, 8:00, 11:00, 14:00, and 20:00 for RO, as well as 2:00, 5:00, 7:00, 10:00–11:00, and 17:00 for SO. Based on the above comparison in the actual scenario, these results also verify the superiority of global interval optimization in efficient and steady performance.

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